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Empowerment and Climate Resilience in Nandi, Kenya: Insights from a Mixed Methods Approach Using Pro-WEAI

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Abstract

This study examines the relationship between empowerment and climate resilience in a mixed crop-livestock system in Nandi County, Kenya, guided by three core research questions: (1) To what extent are women (and men) engaged in agriculture in Nandi County, Kenya, empowered, and what are the key areas of disempowerment? (2) Does empowerment vary by value chain? (3) What is the relationship between empowerment and capacities to cope with and adapt to climate-related shocks?

Using the project-level Women's Empowerment in Agriculture Index (Pro-WEAI) alongside qualitative data from focus group discussions and interviews, the study adopts a mixed methods approach to provide both standardized measures and context-specific insights. Findings reveal women and men are similarly disempowered in this context, with only 9% of women and 18% of men meeting the 80% adequacy threshold. Inadequacy was observed across indicators of self-efficacy, autonomy in income, and control over use of income, with women having greater inadequacy in work balance and visiting important locations compared to men.

Qualitative insights enriched understanding of how empowerment is locally defined and experienced, with women emphasizing independence, financial stability and decision-making, while men highlighted economic capacity and ability to support others. Despite some convergence between the quantitative and qualitative findings—such as the importance of group membership—inconsistencies emerged, particularly around ownership of land and other assets and access to and decisions on financial services. Measures of empowerment varied only slightly by value chain, with persistent intra-household gender gaps.

In relation to climate resilience, a statistically significant association was found between empowerment and whether survey participants changed their practices in response to climate shocks. Those who are empowered are less likely to change their practices, which may indicate greater absorptive capacity to cope with and buffer from immediate effects of shocks. Similar findings emerged for livestock keepers who grow tea and horticultural crops. Their diversified portfolio in a mixed crop-livestock system may have served as an internal cushion, reducing the urgency to change practices in the aftermath of a shock. Future research using longitudinal data and mixed methods approach would be important for a deeper understanding of the causal mechanisms linking empowerment and resilience.

1. Introduction

1.1 Background and rationale

Climate change can intensify already existing vulnerabilities at the intersection of gender, poverty, ethnicity, age, or geographic location. These factors interact to shape everyday decision-making around climate change adaptation (Kaijser and Kronsell, 2014). In agri-food systems, gender differences in access to resources, services, networks and technologies and control over assets, underpinned by patriarchal norms, limit women's voice and agency and shape their resilience capacities (Bryan et al., 2024). For instance, evidence from climate change hotspots across Asia and Africa reveal women may often resort to coping strategies that undermine their well-being and resilience outcomes (Rao et al., 2019). Adaptation processes can be uneven, and unless structural inequalities are addressed as root causes, they can contribute to maladaptation (Schipper, 2022) or reinforce differential vulnerabilities (Eriksen, et al., 2021).

Despite calls for research on approaches that go beyond reaching and benefiting to empowering women and reducing gender inequalities, evidence remains limited, especially through a gender and climate lens (Bryan et al., 2024). Women's empowerment plays a critical role in enhancing household and community resilience to climate shocks. Empowerment can be conceptually defined as a process of change that enhances people's ability to make strategic life choices, particularly in situations where this ability had been denied to them (Kabeer, 1999). In Kabeer's (1999) definition, empowerment consists of 3 dimensions: *resources* (not only access but also future claims), *agency* (including but not limited to decision-making), and *achievements* (outcomes). Out of these three, agency remains the most elusive to measure (Quisumbing et al., 2023). While previously, empowerment was widely seen as a pathway to achieve developmental outcomes related to nutrition, health, or productivity, it is now considered a goal for agri-food systems interventions (Elias et al., 2021).

Literature identifies a notable gap in research on empowerment and outcomes related to environment and natural resources (Quisumbing et al., 2023). This ties in with the need for more research on the relationship between empowerment and climate resilience. Empowered women may be more likely to better anticipate and absorb effects of climate shocks and participate in decision-making around adaptive practices. Abilities to do so could also be unevenly distributed across livelihoods and value chains. Assessing women's and men's empowerment in a certain context—and how it interacts with climate resilience—is essential for addressing the knowledge gap and designing inclusive, evidence-based interventions for sustainable and equitable resilience outcomes. Applying standardized tools and indicators to measure empowerment would be necessary here to contribute to the understanding of “which approaches work, under which conditions, and in which contexts” (Bryan et al., 2024: 9).

Since 2010, there has been much progress on measuring women's empowerment in agricultural contexts through tools and metrics that evolved from treating empowerment as unidimensional to those that recognize empowerment as multilevel and multidimensional (Quisumbing et al., 2023). Launched in 2012, Women's Empowerment in Agriculture Index (WEAI) measures women's agency in agriculture across five domains—production, resources, income, leadership, and time—based on survey responses from women and men

in the same households. WEAI consists of 2 sub-indices: 1. the Five Domains of Empowerment index (5DE) measuring women's empowerment at the individual level, and 2. the Gender Parity Index (GPI) that compares empowerment of women and men within the same households.

Zaremba et al. (2024) observes that empowerment is inherently context-specific and varies widely across individuals and cultures. Focusing narrowly on the empowerment of a single group, such as women, also risks overshadowing or neglecting the disempowerment faced by other marginalized groups. Integrating a broader understanding of empowerment as a multidimensional, relational, and intersectional process would therefore be critical.

To expand the scope of WEAI and understand any potential unintended consequences such as backlash from men and women's time burdens that can then negatively affect outcomes, Project-level WEAI (Pro-WEAI) was developed, sharing a core set of common indicators with WEAI (Malapit et al., 2019; Quisumbing et al., 2023). Pro-WEAI builds on and adapts WEAI, both as a diagnostic tool and as metric to measure impact of interventions on women's empowerment (Malapit et al., 2019). This study applies the Project-level Women's Empowerment in Agriculture Index (Pro-WEAI) in Nandi County, Kenya, mainly as a diagnostic tool to assess empowerment and key areas of disempowerment among women and men in a mixed crop livestock system and to explore how empowerment relates with capacities to cope with and adapt to climate-related shocks.

Drawing on Malapit et al. (2019), a Pro-WEAI survey protocol was used to measure empowerment, along with perceptions of climate impacts and practices related to coping with climate shocks. Findings from quantitative analysis of the survey data were complemented by qualitative insights from interviews and focus group discussions. Triangulating insights from multiple sources allowed for integration of an "etic" (outsider) view from the quantitative analysis with "emic" (insider) perspectives on empowerment emerging from the qualitative approach. Informed by the gaps in literature, our study contributes to a growing body of knowledge on empowerment and resilience capacities (e.g., see Bryan et al., 2024) by combining the strengths of qualitative and quantitative approaches through mixed methods design. We ask the following questions:

1. To what extent are women (and men) engaged in agriculture in Nandi County, Kenya, empowered, and what are the key areas of disempowerment?
2. Does empowerment vary by value chain?
3. Is there a relationship between empowerment and capacities to cope with and adapt to climate-related shocks?

1.2 Conceptual framework

Domains of agency

The project-level Women's Empowerment in Agriculture Index (Pro-WEAI) is anchored in Kabeer's framework of empowerment, and measures agency as one of the key empowerment dimensions (Kabeer, 1999). Unlike the original WEAI with five domains of empowerment and 10 indicators, Pro-WEAI maps 12 indicators to three domains of agency: intrinsic agency (power within), instrumental agency (power to), and collective agency (power with). These three domains reflect a typology of generative forms of power (Rowlands, 1997;

Ibrahim & Alkire, 2007), with the exception of coercive aspect or “power over” which is not included in the index (Malapit et al., 2019).

Intrinsic agency

There are four indicators of intrinsic agency. They include autonomy in income, self-efficacy, attitudes about intimate partner violence (IPV) against women, and respect among household members (Malapit et al., 2019). **Autonomy in income** measures autonomy in how an individual spends his or her income from agricultural and non-agricultural activities. **Self-efficacy** measures participants’ perceptions of their capabilities to achieve goals and pro-WEAI does this by using the New General Self-Efficacy Scale (NGSE). The scale has statements with scores for how strongly participants agree or disagree. This indicator was not included in the original WEAI. **Attitudes about IPV against women** indicator examines participants’ attitudes towards intimate partner violence against women in multiple scenarios. This indicator was not included in the original WEAI. **Respect among household members** evaluates if there are respect and trust among the participant and spouse or another household member (of the opposite sex) under certain conditions, including whether the participant feels comfortable disagreeing with the household member. This indicator was not a part of the original WEAI.

Instrumental agency

Instrumental agency is measured using six indicators, including input into productive decisions, ownership of land and other assets, control over use of income, access to and decisions on financial services, work balance, and visiting important locations. **Input into productive decisions** measures the contribution of participants in decision making for all the agricultural activities they participate in. It examines whether individuals make decisions solely or jointly with some input in the decisions made, and if they feel like they could participate in decision making if they wanted to. **Ownership of land and other assets** looks at what is owned, either solely or jointly, and goes further to assess ownership of small and large assets including land in the household. Adequacy is achieved when an individual in the household solely or jointly owns at least three movable assets or land. Households that do not own any assets are considered inadequate in this indicator. **Access to and decisions on financial services** indicator examines whether participants have access, solely or jointly, to financial accounts, if they have borrowed money and made decisions on that jointly or solely in the past one year, or could have borrowed from at least one source of credit if they wanted to. **Control over use of income** measures input in decisions regarding the use of income generated from activities the household members participate in, including agricultural and non-agricultural activities. **Work balance** indicator uses 24-hour recalls distinguishing between time spent on paid and domestic or caregiving work. The indicator measures time spent on primary activities (work related tasks) and secondary activities (childcare) to determine the adequacy. Individuals are adequate in this indicator if they work for less than 10.5 hours (WEAI poverty line) in the past 24 hours. **Visiting important locations** examines the ability to be mobile by finding out how often they visit different locations on a weekly or monthly basis, including urban centers, markets, family/relatives, public meetings and health facilities. This indicator was not included in the original WEAI.

Collective agency

Collective agency is comprised of 2 indicators—group membership and membership in influential groups. **Group membership** requires the participant to be active in at least one group. For communities that do not have groups, participants are considered inadequate in

this indicator. Being an active member of a group could mean paying membership fees if any, holding leadership positions, frequently attending meetings, among others. **Membership in influential groups** assesses if the participant is an active member of at least one group that can influence the community to a certain extent. This indicator was not a part of the original WEAI.

Resilience capacities

Climate resilience refers to “the capacity of interconnected social, economic and ecological systems to cope with a hazardous event, trend or disturbance, responding or reorganizing in ways that maintain their essential function, identity and structure. Resilience is a positive attribute when it maintains capacity for adaptation, learning and/or transformation” (IPCC, 2023). Resilience is not a static state of being, but a set of capacities that enable effective individual and collective responses to both sudden shocks and gradual stresses. Resilience capacities can be understood in relation to the timeframe in which they are operationalized. The 3As framework of resilience (see Bahadur et al., 2015) focuses on three interrelated capacities: **absorptive capacity**, **anticipatory capacity**, and **adaptive capacity**. These capacities can be defined as follows:

Absorptive capacity

The ability to absorb and cope with the immediate impacts of climate extremes and variability. This is essentially a system’s short-term coping capacity exercised during or after a disturbance. For example, having savings, insurance, social support networks or indigenous coping practices can help households absorb the impact of shocks by buffering against losses and meeting basic needs during crisis.

Anticipatory capacity

The ability to anticipate and reduce the impact of potential shocks through preparedness and early action. This is a *proactive, preventive* capacity, encompassing preparedness measures, early warning systems, and forward-looking risk management.

Adaptive capacity

The ability to adjust, learn, and adapt in the face of long-term changes or future shocks. This is a system’s capacity for gradual adjustment through innovation or new strategies. In practice, adaptive capacity might involve diversifying livelihoods, adopting new climate-smart technologies, or modifying institutions and behaviors to better suit a changing climate.

Notably, some frameworks also discuss *transformative capacity*, defined as the “ability of the system to change its own structures” and address root causes of vulnerability (Castillo et al., 2017). Transformation can also be seen as an overarching process that enhances all 3As rather than a separate capacity. As a broader concept, transformation requires engagement with power relations and structures that influence decision-making (see Bahadur et al., 2015) and is outside the empirical scope of this study.

2. Methodology

2.1 Study site

Nandi County is located in Kenya's North Rift region and covers an area of approximately 2,884 square kilometers. The county experiences annual rainfall averaging 1,200–2,200 mm and temperatures ranging from 12°C to 27°C. Rainfall follows a bimodal pattern, with long rains starting from March to June and short rains from September to October. Impacts of climate variability in the form of erratic rainfall result in flash floods and occasional droughts threaten soil health, water resources, and agricultural productivity. The county is administratively divided into six sub-counties (Emgwen, Chesumei, Mosop, Nandi Hills, Tinderet, and Aldai) and a total of 30 wards. According to the Kenya National Bureau of Statistics (Kenya National Bureau of Statistics, 2019), Nandi County had a population of 885,711, with almost 90% living in rural areas, reflecting the county's agricultural orientation. Approximately 36.1% of the population lives below the poverty line.

Agriculture forms the backbone of Nandi's economy. Out of 148,660 farming households, 96.2% engage in crop farming, while 72.9% are involved in livestock farming. Specifically, 40,206 households practice crop-only farming, 5,505 are livestock-only, and 102,872 combine both activities. 70.4% of the households are male-headed compared to 29.6% female-headed households. Most farmers are smallholders, cultivating 1.2–1.5 hectares on average under mixed farming systems. The agriculture practiced is predominantly rain-fed and mixed farming. Major value chains include dairy, maize, tea, coffee, and legumes, which contribute significantly to household incomes and food security. The dairy sector is particularly vibrant, supported by cooperative societies and county investments in processing infrastructure such as the Kabiye Dairy Plant, poultry facilities, and milk cooling and bulking centers (County government of Nandi, 2023).

Infrastructure development remains pivotal for market access. The county has 11 open-air markets, and moderately well-connected road network. Poor road conditions, particularly in hilly and rain-prone areas, continue to hinder efficient transport. Recent investments in road rehabilitation, market modernization, electricity expansion, and processing facilities have improved connectivity and reduced post-harvest losses. However, water and irrigation infrastructure remain underdeveloped, leaving most farmers reliant on rain-fed agriculture.

Various institutions play an essential role in supporting smallholder farmers and promoting sustainable agricultural development. The Nandi County Department of Agriculture and Livestock provide extension services and training. Cooperative societies, such as the Nandi Dairy Cooperative Union and local tea and coffee groups, facilitate access to markets, inputs, and financial credit. Additionally, development partners and non-governmental organizations, including programs like the Agricultural Sector Development Support Programme (ASDSP II), the Kenya Climate-Smart Agriculture Project (KCSAP) support initiatives in value chain strengthening, agribusiness capacity building, and climate adaptation. Community-based and self-help groups, especially women-led savings groups and associations foster financial inclusion and collective action at the grassroots level.

The county is predominantly inhabited by the Nandis, a sub-group of the Kalenjin community. Gender norms strongly influence and are influenced by agriculture as a dominant livelihood. Women provide substantial on-farm labor, particularly when it comes to caring for livestock, yet they have limited access to resources like land and finance, and agency to make

decisions. Men typically control income from crops and livestock. Women's membership in self-help groups can support economic empowerment, with increased access to inputs, finance, and extension services. While patriarchal norms around ownership of land and livestock shape intra-household relations and decision-making on adaptation that are largely inequitable, interaction with factors like age or generation, household arrangements, and socio-economic conditions may help explain emergent positive deviations from restrictive norms and practices (Bullock et al., 2025).

2.2 Sampling and data collection

Quantitative data collection

A multistage sampling strategy was employed to facilitate comparative analysis across Nandi County. All 30 wards in the county were stratified based on agroecological conditions and dominant agricultural activities to ensure representativeness and internal homogeneity within each stratum. Tea, livestock keeping, and horticulture were identified as the main value chains. These value chains were distributed across three distinct agroecological zones, characterized primarily by differences in altitude and precipitation. The final sample was randomly selected from 15 wards across six sub-counties, ensuring adequate representation across locations and value chains (Table 1).

The pro-WEAI survey team consisted of 8 enumerators, 4 women and 4 men, and one team leader. Survey data was collected in partnership with Kaimosi Agricultural Training Center (KATC). Ethical approval was provided by ILRI's Institutional Research Ethics Committee (IREC) as well as by the Kenya National Commission for Science, Technology and Innovation (NACOSTI).

Table 1: Wards and sub-counties in the final sample

Sub County	Ward
Aldai	Kaptumo/Kaboi
	Terik
	Kemeloi
	Koyo/Ndurio
Chesumei	Chemundu/Kapng'etuny
	Kosirai
Emgwen	Kapkangani
	Kilibwoni
Mosop	Kabiyet
	Kipkaren
	Ndalat
	Sangalo/Kebulonik
Nandi Hills	Chepkunyuk
	O'llessos
Tinderet	Songhor/Soba

A total of 472 participants were surveyed from 253 households. Following Pro-WEAI guidelines, two adult members (a man and a woman) were interviewed per household where possible. In households with only one adult, the survey had to be conducted only if this

person was a woman. Data was collected using the Open Data Kit mobile data collection tool. The final dataset used in the assessment of empowerment (Pro-WEAI analysis) consisted of 250 women and 219 men. Households without a female participant where only men were interviewed were excluded from the final Pro-WEAI calculations (Table 2).

Table 2: Sample size for Pro-WEAI analysis

Household type	Number of men interviewed	Number of women interviewed	Total number of observations (n)
Women only (female-headed)	0	32	32
Men only (single-male adult households)	2	0	2
Mixed (male and female) household	219	219	438
Total number of observations	221	251	472
Total number of observations used in Pro-WEAI analysis	219	250	469

Note: Two men from households with no women participants were excluded. One woman participant had data for 2 indicators missing and had to be removed from the final sample used in the Pro-WEAI analysis (n=469).

The following modifications were made to the Pro-WEAI survey protocol:

- 'Horticultural crops' in module G2 was separated into 'tea farming and processing of harvest' and 'other horticultural crops farming' to factor in value chains.
- A question on how the participants acquired their land was added in module G3.01.
- A question on whether they have childcare facilities and if they would or have used the facilities was added in module G4.
- An additional module on 'life satisfaction'—G8 (C) was added.
- Only section D of module G8 (A) was retained.
- A module on climate change was added with three submodules on climate shocks, climate change perceptions, and mitigation strategies (Module 13).

Value chain groups

Two distinct value chain groups could be identified from the final sample: 1. Livestock keepers who did not grow tea; 2. Livestock keepers who grow tea and horticultural crops. A small percentage of the sample (2.6%) did not belong to any of these 2 groups i.e., neither livestock keepers nor livestock keepers who are also tea/horticulture farmers (Table 3) and were thus excluded from part of the analysis that focused on differences between these value chain groups.

Though these can be conceptualized as separate value chains, the sample could not be neatly divided into distinct groups of livestock keepers, tea growers and horticulture farmers. The overlap present here highlights unique characteristics of a mixed crop-livestock system where livestock keepers would also often grow crops on their land (see Annex 1 for detailed demographics of the survey participants).

Table 3: Value chain groups in the sample

Value chain groups	Number of participants	% of participants	Female only households	Mixed households
Livestock keepers only	239	51%	5%	95%
Livestock + Tea + horticulture farmers	218	46%	8%	92%
Neither livestock keepers nor tea/horticulture farmers	12	3%	0%	100%
Total number of observations used in Pro-WEAI analysis	469	100	31	438
Total number of observations used in Pro-WEAI (VC)analysis	457		31	426

Qualitative data collection

Qualitative data collection instruments were drawn and adapted from the GAAP2 PRO-WEIA qualitative protocols – the semi-structured life histories interview – as well as GENNOVATE's 'capacity for innovation' tool (IFPRI 2018; Petesch et al., 2018). We conducted four sex-disaggregated focus group discussions (FGDs) with six to eight participants each following the 'capacities for innovation' FGD guide (see Annex 2).

A total of 29 women and men participated in the FGDs with a minimum of six and maximum of eight participants per group. Two of the women participating were single at the time of the study, while all others were married. Majority of the participants were below the age of 47 (Table 4.1, see Annex 2 for detailed demographics of the focus groups). The focus groups explored perceptions and experiences of empowerment, gender norms and practices, division of labor, access to networks, and control over productive assets in addition to opportunities and barriers for uptake of innovation.

Table 4.1: Demographic characteristics of FGD participants in Nandi

Location (Ward)	Gender	Number of participants	Age range	Highest Education
Kaptumo	Women	6	45-60	Form 4
Kapkangani	Men	8	28-71	College
Kosirai	Men	7	35-70	College
Kipkaren	Women	8	16-57	College

Table 4.2: Demographic characteristics of interview participants in Nandi

Location (Ward)	Sex	Age	Marital status	Highest Level of Education Completed	Length of time lived in the community (years)
Kipkaren	Woman	45	Married	Form 3	25
Kipkaren	Woman	38	Single	Form 4	38
Kapkangani	Man	60	Married	Primary	54
Kosirai	Man	57	Married	Post graduate	57
Kosirai	Man	76	Married	Form 2	76
Kapkangani	Man	62	Married	University degree	53
Kaptumo	Woman	34	Married	Bachelors degree	9
Kaptumo	Woman	33	Married	Diploma	3

An additional eight semi-structured interviews following the ‘life histories’ guideline were carried out to examine perceptions and experiences of women’s empowerment in Nandi, and to validate elements within the pro-WEAI survey tool (Table 5). Amongst the life history interview participants, the youngest participant was 33 at the time of the study, and the oldest 76. Multiple participants had moved to the communities they now live in after they got married. The youngest women, aged 34 and 33, had only lived in the community 9 and 3 years respectively. It is also worth noting that these young women both received higher education, beyond high school (Table 4.2).

Qualitative data was collected in the same sub-counties where the Pro-WEAI survey was conducted. But while the survey collected data from participants across the different wards of each sub-county, qualitative data was only collected in one ward per sub-county due to budget and time limitations. Selected wards, based on socio-economic and agro-ecological diversity were Kaptumo-Kaboi in Aldai sub-county, Kosirai in Chesumei sub-county, Kapkangani in Emgwen sub-county, and Kipkaren in Mosop sub-county.

Table 5: Qualitative data collection instruments

Tool	Thematic focus	Number of participants
Life history interviews	Life history interviews focused on perceptions and experiences, and how women and men’s decision-making, time use, and group participation, among other indicators, have changed over time.	8 (4 women; 4 men)
Focus group discussions	The focus group discussions aimed to capture perceptions of empowerment and understand agency, specifically in relation to women’s roles, decision-making in production, and changes in practices in response to climate shocks.	29 (14 women; 15 men)

2.3 Analytical framework

Quantitative analysis: Measuring empowerment

Pro-WEAI consists of the following 12 indicators to measure empowerment dimensions across three domains of agency:

- **Intrinsic Agency:** autonomy in income, self-efficacy, attitudes toward intimate partner violence, respect among household members.
- **Instrumental Agency:** input in productive and income decisions, ownership of land and assets, access to and decisions on financial services, control over income use, work balance, visiting important locations.
- **Collective Agency:** group membership, membership in influential groups.

All 12 indicators were used in this study and weighed equally in the index (each contributing 1/12 of the total). Each indicator has a defined adequacy threshold that determines whether an individual's achievement in that dimension is sufficient (i.e. "adequate"). For example, a person is considered *adequate in group membership* if they are an active member of at least one community group. An individual's empowerment score is thus the weighted average of their adequacy across these 12 indicators. Using the Pro-WEAI criteria, an individual is considered empowered if they achieve adequacy in at least 80% of the indicators. Falling below this threshold classifies the individual as disempowered.

Calculation of the index involves 2 parts. The first is the Three Domains of Empowerment (3DE), which reflects both how many individuals are empowered and the extent of empowerment across the three domains. Empowerment status was assigned to participants who achieved adequacy in 80% or more of indicators. The second is the Gender Parity Index (GPI), which assesses women's empowerment relative to men in the same households. GPI provides a score for women only. This index is calculated at the household level, and a household achieves gender parity if the woman's empowerment score is at least as high as the man's *or* if the woman herself is empowered according to the Pro-WEAI criteria. A GPI of 1 or close to 1 reflects parity between women and men within households.

Box 1: Components of Pro-WEAI

Percent achieving empowerment refers to proportion of participants in the sample who are empowered, also referred to as the adequacy headcount ratio.

Percent not achieving empowerment (H): the number of individuals in the sample who are disempowered, i.e., they fail to demonstrate adequacy in at least 80% of the indicators. This is also referred to as the inadequacy headcount ratio.

Mean disempowerment score (A) indicates the intensity of disempowerment for men and women. It is calculated as the mean of indicators in which disempowered women and men are inadequate.

Percent not achieving gender parity (HGPI) refers to the proportion of households in the sample that do not achieve gender parity, or in which the disempowerment score of the woman is greater than that of the man, and the woman is not empowered.

Percent achieving gender parity refers to the proportion of households in the sample that achieve gender parity, or in which the empowerment score of the woman is greater than or equal to that of the man.

Mean empowerment gap is the average percentage shortfall that a woman without parity experiences relative to her spouse. It is only calculated for households without gender parity and reflects average difference between the empowerment scores of a man and a woman in a household.

The overall Pro-WEAI score for a given sample or project is computed as a weighted composite of the 3DE and GPI. In the standard formulation, the 3DE contributes 90% and the

GPI 10% to the combined index. 3DE reflects the extent and intensity of individual empowerment in the sample, while GPI gives the extent and intensity of gender parity in the sample. The Pro-WEAI score is $= 0.9*(3DE) + 0.1*(GPI)$, yielding a number between 0 and 1. A higher Pro-WEAI value indicates greater overall empowerment in the sample, factoring in both women's achievements across all 12 indicators and the degree of parity between women and men in households. Additionally, the findings were visualized through bar graphs to illustrate inadequacy patterns across indicators and understand key contributors to disempowerment.

Quantitative analysis also aimed to understand how empowerment might relate to resilience capacities, and whether capacities to cope with and adapt to climate-related shocks differ by value chain. We used Pearson's chi-squared test of independence to assess if there is an association between climate shocks and changes in practices for those empowered and not empowered, and the 2 groups of livestock keepers and livestock keepers who are also tea/horticulture farmers.

Qualitative Analysis

Coding of qualitative data mainly followed a deductive approach. Themes identified from the data broadly aligned with the thematic focus of the data collection instruments. Thematic analysis was organized around perceptions of empowerment, gender norms and decision-making in households, experiences related to the three domains of agency, key constraints and barriers to empowerment, as well as resilience capacities. Inductive themes also emerged from the data, including changes in norms and practices and collective agency.

Mixed methods approach

Quantitative results were accompanied and to some extent partially explained by interpretation of qualitative data. This approach integrating the two methods enabled drawing of meaningful inferences from the data. Through triangulation of insights from multiple sources, we note where qualitative findings support the quantitative results, and where they depart from each other. Combining insights from quantitative and qualitative findings allowed us to bring "etic" (outsider) and "emic" (insider) perspectives together for a deeper context-specific understanding of empowerment and resilience.

3. Key findings

3.1 Overview of empowerment in Nandi

Applying the 80% adequacy threshold, only 9% of women and 18% of men in the sample are classified as empowered. This means that 91% of women and 82% of men are disempowered. Overall, women's average 3DE score was 0.61 and men's 0.67. Among disempowered women, the mean 3DE was 0.57, and 51% of dual adult households in the sample did not achieve gender parity. The Gender Parity Index (GPI) was 0.87 which is less than 1, reflecting a gap in parity between women and men within households. The overall composite Pro-WEAI score was 0.64 (Table 6), which is a weighted average of GPI and 3DE, with 3DE carrying 90% of the weight and GPI 10% of the weight.

Within households, 53% of the sample had men with greater adequacy scores compared to women. Women with greater adequacy scores compared to men in households made up

25% of the sample, while less than one fourth of the sample had women with equal adequacy scores as men (see Table 7). At an individual level, a major proportion of the sample is comprised of women and men who are disempowered.

Table 6: Pro-WEAI results

Pro-WEAI		
Indicator	Women	Men
Number of observations	250	219
3DE score	0.61	0.67
% achieving empowerment	9	18
% Not achieving empowerment (H)	91	82
Mean 3DE score for not yet empowered	0.57	0.6
Mean disempowerment score (A)	0.425	
Gender Parity Index (GPI)	0.87	
% achieving gender parity	49	
% Without gender parity (HGPI)	51	
Average intra-household inequality score	0.05	
Empowerment gap	0.25	
Pro-WEAI	0.64	

Table 7: Differences in adequacy scores among women and men

Difference in adequacy scores	Percent
Male adequacy score > female adequacy score	52.5%
Female adequacy score > male adequacy score	24.7%
Female adequacy score = male adequacy score	22.8%
Total	100.0%

“Emic” perspectives on empowerment

Qualitative findings highlight “emic” views on empowerment. Initially, instead of using the term ‘empowerment’, the focus group participants were asked to share qualities about someone they look up to, what his/her life is like, and what he/she may have achieved. Women noted that they look up to or appreciate individuals who are hardworking, have vision, are committed and good time managers, therefore emphasizing personality traits over asset ownership. In contrast, men shared that they respect or look up to individuals who have capital, can cater for the welfare of the family and are good at farming. When asked specifically what empowerment means to them, men highlighted economic benefits, higher educational attainment, and the ability to support others, while women often associated the term with independence, financial stability and decision-making power. According to the women participating in focus group

discussions, empowerment relates with improved livelihood, food security and future aspirations.

In a focus group of men, one participant noted, *“An empowered woman has money and knowledge; she can help others and live a good life”* (R3). Similarly, women participants highlighted that an empowered individual is one who not only supports themselves but also uplifts their family and community. This underscores a common perspective that empowerment brings a duty to support others. Empowerment is therefore perceived beyond an individual and extends to household and community.

Despite predominantly positive perceptions of empowered women, it was also clear that being empowered can come with disadvantages. Empowered women can be both admired and envied by others and even face backlash from men in their household and community. As highlighted by women in FGD 4:

“Men see it as overpowering her husband. Young boys can even steal from you thinking you are now wealthy” (R4)

While participants commonly described an empowered woman as someone who have and can control their own money, it is worth noting here that none of the participants in the qualitative study considered themselves empowered, often citing lack of capital as a major constraint.

3.2 Inadequacy by indicators

Uncensored headcount ratio, reflecting the proportion of women and men in the sample who are inadequate in the indicator, regardless of whether they are empowered or disempowered, revealed lower inadequacy in ownership of land and other assets, access to and decisions on financial services and group membership (Table 8).

Qualitative findings align with lower inadequacy in group membership contributing to the domain of collective agency. This highlights the importance of groups or collectives for overcoming individual constraints to empowerment in this context. Participants, especially women, shared about their involvement in community-driven financial initiatives, such as informal savings and credit groups. These groups allow members to pool resources and provide support through table banking approaches. As shared by an interview participant:

“I joined Joywo group to save and qualify for a loan, which has helped pay school fees” (woman, 45 years old, Kaptumo).

Despite the Pro-WEAI analysis showing lower inadequacy in indicators related to ownership of land and other assets and access to and decisions on financial services, qualitative findings reveal a more complex reality. While men acknowledged land-related challenges, their concerns focused more on limited land size for experimenting with new practices and the absence of critical infrastructure, such as irrigation systems. Women described structural barriers that limit their control over land and financial resources. FGD participants emphasized cultural norms often prevent women from independently accessing or using land, and that decisions about land must be negotiated with their husbands. A recurring theme was that land title deeds are typically registered in men’s names, reinforcing women’s lack of formal ownership.

Thus, while access to financial resources were reported by both women and men as a key constraint, for women, however, limited access to finance is compounded by restricted decision-making power. Even when women have some access to financial services—whether through formal channels or informal savings groups—the ability to control and make independent financial decisions is often constrained by cultural norms. As noted by one women’s FGD participant, “Women don’t have money of their own so can’t make decisions. It is a cultural issue” (R3). This highlights the need to not only ensure there is availability and equitable access to resources but also agency to make decisions around how they are used.

Table 8: Inadequacy by indicators

	Indicators	Women	Men
<i>Intrinsic Agency</i>	Autonomy in Income	56	65.8
	Self-efficacy	63.2	52.1
	Attitudes about IPV against women	36	26.9
	Respect among household members	47.6	32.4
<i>Instrumental Agency</i>	Input in household decisions about production and income	31.6	37.4
	Ownership of land and other assets	1.6	2.28
	Access to and decisions on financial services	8	4.11
	Control over use of income	49.6	62.1
	Work balance	64.8	31.1
	Visiting important locations	57.2	45.2
<i>Collective Agency</i>	Group membership	20.4	25.1
	Membership in influential groups	44.8	37

Gender differences in inadequacy were noted. Prominently, women are inadequate in work balance and self-efficacy among other indicators, while for men, there is notable inadequacy in autonomy in income and control over use of income. Women’s inadequacy in work balance is further supported by qualitative findings. Women report spending many hours on domestic tasks, such as cooking, cleaning, and childcare, which limits the time they can allocate to income-generating activities. Women focus group participants noted time availability as a key constraint to innovation. “We don’t have time and always need to adjust. We need to manage the household. Men have more time” (R23). When speaking about the barriers that one woman faces to achieving her goals in life, one interview participant said, “Women are seen as not having time due to children and household chores” (woman, 33 years old, Kaptumo). This has implications for women’s time in income earning activities. For instance, as mentioned by men in one focus group: “women don’t have time for their other duties if they are earning income” (R11). Overall, key contributors for both women’s and

men's disempowerment included limited control over use of income, low autonomy in income, weak self-efficacy, limited ability to visit important locations and low membership in influential groups (Figure 1).

For collective agency, the indicator group membership has lower inadequacy for both women and men compared to the indicator membership in influential groups. This signifies that group membership, relative to membership in influential groups, contributes less to disempowerment and can thus play a key role in strengthening collective agency. However, membership in groups may not automatically translate to influence.

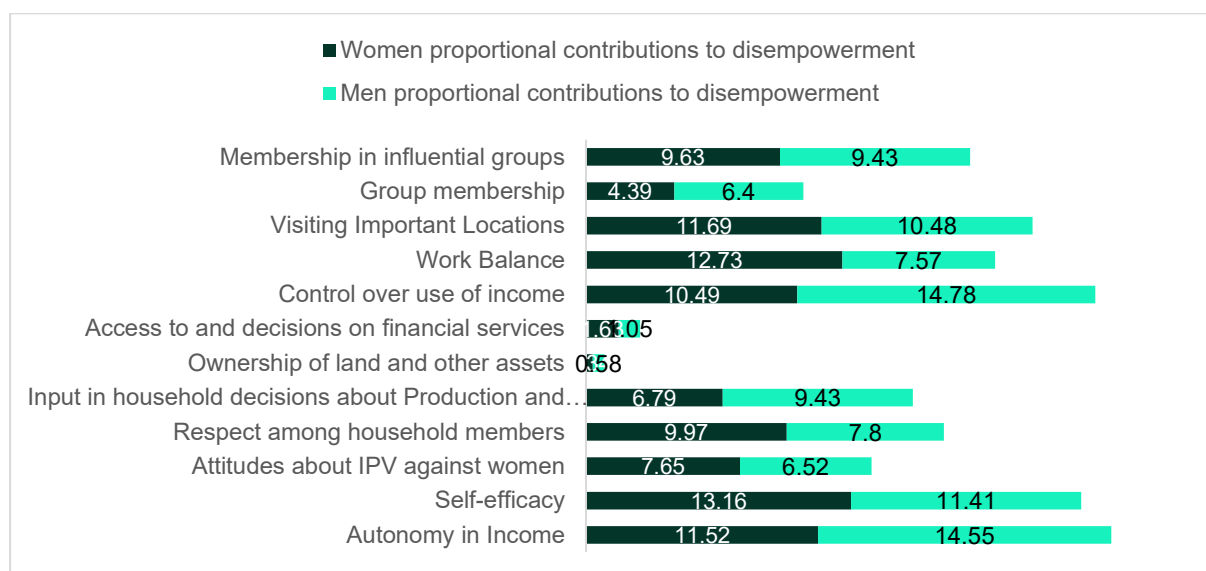


Figure 1: Key contributors to disempowerment (source: Pro-WEAI analysis)

Qualitative findings also demonstrate women's involvement in groups. Groups and networks are critical mechanisms that support knowledge sharing and pooling of resources. Those who participate in local associations or organizations, such as church groups and savings cooperatives, describe a sense of belonging and shared purpose. Participants described that women often bring new ideas and engage actively in group activities, particularly through table banking initiatives and interactions with extension as a group. But financial barriers, such as the inability to meet monthly contributions, can lead to women periodically withdrawing from groups. As noted by an interview participant, "I left because I didn't have enough income, I now have my poultry and can go back" (woman, 33 years old, Kaptumo). Additional concerns include group members defaulting on payments, which place strain on others, and feelings of embarrassment or low self-confidence that discourage attendance.

3.3 Empowerment by value chain groups

Pro-WEAI analysis by value chain reveals that among survey participants who are only livestock keepers, only 8% of women and 14% of men are empowered, while for those who keep livestock and cultivate tea and horticultural crops, slightly more women (10%) and men (20%) are empowered. The mean 3DE score is 0.57 for disempowered women who keep livestock, and 0.58 for disempowered women who participate in multiple value chains. GPI is 0.87 and the final Pro-WEAI score is 0.63 for those only keeping livestock. GPI is 0.86 and the final Pro-WEAI score is 0.64 for those keeping livestock and growing tea and horticultural crops (Table 9).

Within households who only keep livestock, 50% have men with greater adequacy scores than women, and only 19% have equal adequacy scores between women and men. In the households of livestock keepers and tea and horticulture farmers, 57% have men with higher adequacy scores than women in the household, and 27% with equal adequacy scores between women and men. Women have higher adequacy scores compared to men in 31% of the households who keep only livestock, and in 16% of the households who are livestock keepers and tea and horticulture farmers (Figure 2).

Table 9: Pro-WEAI results by value chain groups

Pro-WEAI	Overall		Livestock		Livestock, tea and horticulture	
Indicator	Women	Men	Women	Men	Women	Men
Number of observations	250	219	128	111	115	103
3DE score	0.61	0.67	0.61	0.65	0.62	0.69
% achieving empowerment	0.09	0.18	0.08	0.14	0.1	0.2
Mean 3DE score for not yet empowered	0.57	0.6	0.57	0.59	0.58	0.61
Gender Parity Index (GPI)	0.87		0.87		0.86	
% achieving gender parity	0.49		0.5		0.46	
Average intra-household inequality score	0.05		0.04		0.07	
Empowerment gap	0.25		0.25		0.25	
Pro-WEAI	0.64		0.63		0.64	

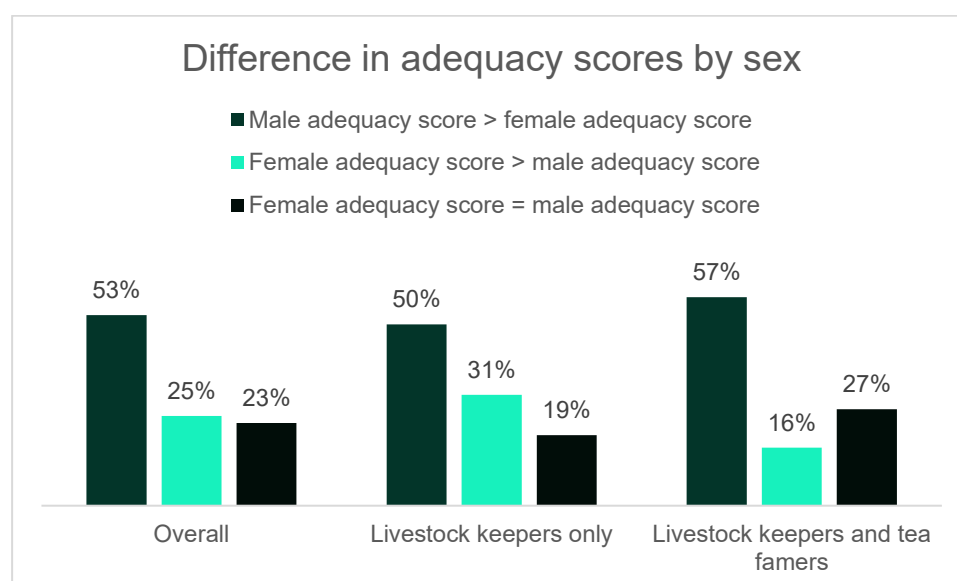


Figure 2: Difference in adequacy scores between women and men within households by value chains (source: Pro-WEAI analysis)

Uncensored headcount ratio for the value chain groups reveals similar patterns in inadequacy as the overall sample. Both women and men have lower inadequacy in ownership of land and other assets, access to and decisions on financial services and group membership (Table 10).

Across the value chains, women have greater inadequacy compared to men in self-efficacy, work balance, visiting important locations, and respect among household members. Men have higher inadequacy in autonomy in income and control over use of income. Notably, women in the livestock only group score higher inadequacy in autonomy in income compared to women in households who keep livestock and grow tea and horticultural crops (Table 10).

Table 10: Inadequacy in indicators by value chains

Indicators		Overall		Livestock		Livestock, tea & horticulture	
		Women	Men	Women	Men	Women	Men
<i>Intrinsic Agency</i>	Autonomy in income	56	65.8	60.9	62.2	51.3	68.9
	Self-efficacy	63.2	52.1	61.7	53.2	63.5	50.5
	Attitudes about IPV against women	36	26.9	36.7	23.4	35.7	30.1
	Respect among household members	47.6	32.4	46.9	36	50.4	29.1
<i>Instrumental Agency</i>	Input in household decisions about production and income	31.6	37.4	32	40.5	31.3	34
	Ownership of land and other assets	1.6	2.28	0	2.7	2.61	0.971
	Access to and decisions on financial services	8	4.11	10.2	7.21	6.09	0.971
	Control over use of income	49.6	62.1	43.8	64	54.8	61.2
	Work Balance	64.8	31.1	64.1	37.8	67	24.3
	Visiting Important Locations	57.2	45.2	62.5	48.6	51.3	41.7
<i>Collective Agency</i>	Group membership	20.4	25.1	21.1	25.2	18.3	25.2
	Membership in influential groups	44.8	37	46.9	41.4	42.6	33

Key contributors to disempowerment by value chain groups follow a similar pattern to the overall sample (Figure 3 and Figure 4). Across the two value chains, the largest contributions to disempowerment for both women and men come from autonomy in income, self-efficacy, control over use of income, ability to visit important locations and membership in influential groups. Compared to men, work balance remains a key contributor to women's disempowerment across the value chains.

Financial decisions around production and sale of large livestock or large-scale farming are typically male dominated, though women focus on specific aspects, such as milking, routine care, and management of casual laborers on farms. This aligns with the findings of the focus group discussions, which identified use of artificial insemination for improved breeds, semi-zero grazing, and livestock feeding practices as the most important for men in Nandi County. Improved poultry breeds and management practices, and new certified vegetable varieties for kitchen gardening emerged as the most important for women. Unlike ownership of cows,

women can own and make decisions around poultry, though the use of income generated from this is still frequently determined by husbands, as articulated by the FGD participants.

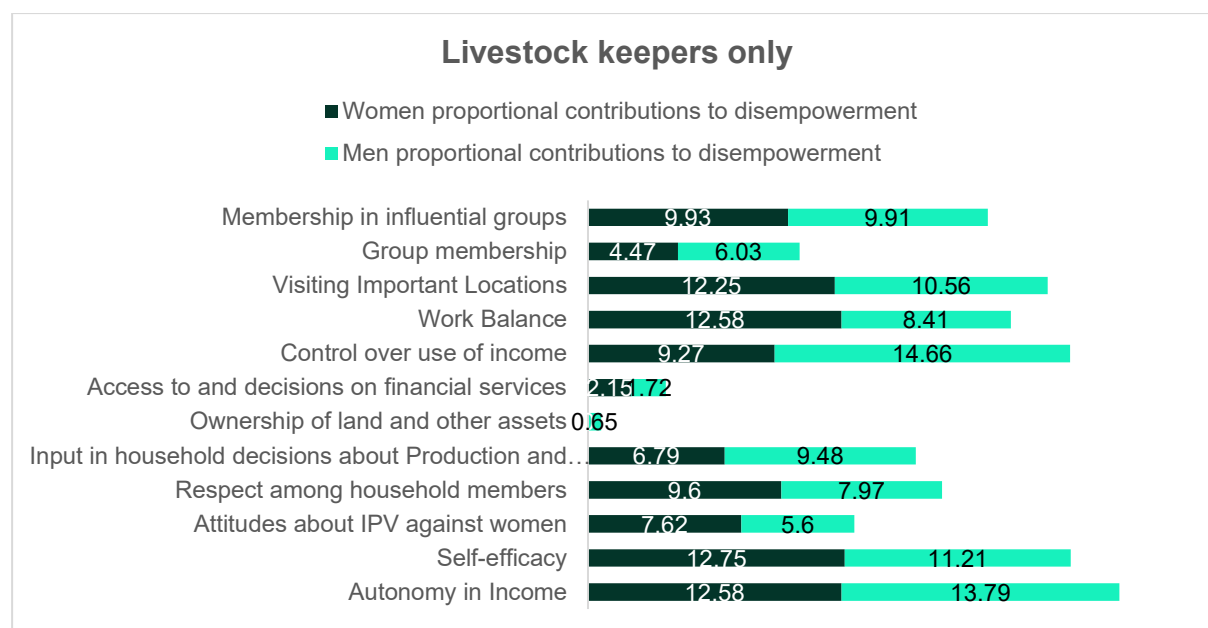


Figure 3: Key contributors to disempowerment for livestock keepers (source: Pro-WEAI analysis)

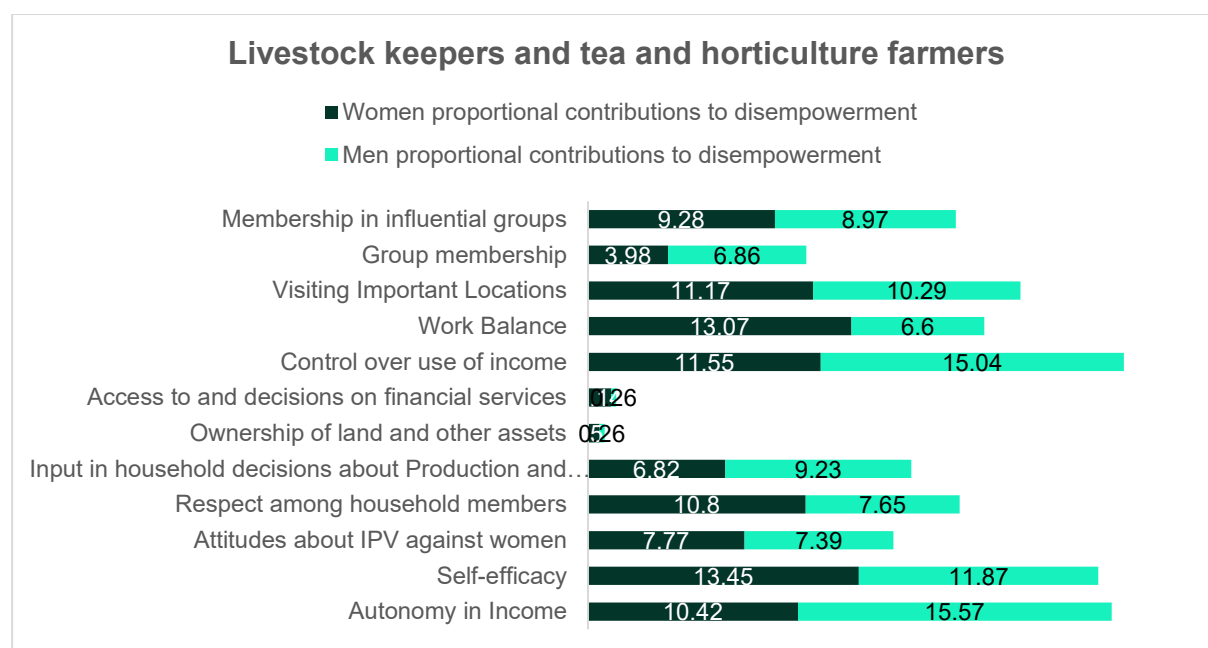


Figure 4: Key contributors to disempowerment for livestock keepers and tea and horticulture farmers (source: Pro-WEAI analysis)

3.4 Empowerment and coping with climate-related shocks

In Nandi, commonly perceived climate shocks and stresses are droughts, floods, and erratic rainfall. Perceptions of key climate-related impacts on households include decline in crop yield, loss of crops, loss of income, and food insecurity. According to an interview participant,

who has been affected by heavy rains and diseases, “This results in low yields, low income and less food for my family” (woman, 38 years old, Kipkaren).

When it comes to coping strategies in response to the climate-related shocks and stresses, there is no significant association between those who are empowered and those who are not empowered. The most common coping strategies reported by these 2 groups were “did nothing”, “bought food/changed eating patterns” and “sale of assets”.

A chi-squared test of independence showed there is significant association for those who are empowered and whether they changed practices in response to climate-related shocks and stresses ($\chi^2 = 15.3$, $df = 4$, $p < 0.05$). Overall, those who are empowered are less likely to change their farming practices in response to climate shocks and stress. This could be indicative of a positive relationship between empowerment and absorptive capacity. Those who are empowered may have improved coping capacity to absorb the effects of shocks over shorter term.

Most of the interview participants did not identify themselves as being able to sufficiently cope with or prepare for future challenges. It was noted that lack of capital or inadequate financial resources is a major constraint, with one participant pointing to livelihood-related challenges.

“We don’t have resources... Only tea provides large income, that’s why so many people are doing it here, but it is at risk... I am thinking of changing from tea... I would plant Boma Rhodes for sale as livestock feed” (man, 62 years old, Kapkangani).

Women poultry farmers reported adapting to harsh weather conditions by taking protective measures. One constructed a poultry house to shield her chickens from heavy rain and wind, while another kept young chicks indoors near the cooking fire to keep them warm.

Additionally, interview participants shared that while there have been some efforts to absorb climate-related impacts, such as pooling resources by conducting communal *harambees* (fundraisers), these strategies are not sufficient to cover large-scale losses. Demand for climate information services also emerged from qualitative data: “I would like more climate information to be able to plan better. I get some from the radio.” (woman, 38 years old, Kipkaren).

Participants also started adopting new practices to cope with the impacts of a changing climate, including using organic fertilizers, planting drought-resistant crops like avocado, and engaging in soil and water conservation techniques such as terracing. Practices like drip irrigation and planting super Napier grass for dry seasons were mentioned.

For those who did report changing practices across the two groups of empowered and not empowered, the most common change occurred in crop related measures, followed by manure and water related measures, and livestock. Diversification and seeking off-farm employment were reported as the least common measure in response to climate shocks and stresses.

There was no significant association between those who are empowered and those who are not empowered when it comes to reporting constraints in response to why they could not change their practices. Across the two groups, no access to money and lack of information on climate change and adaptation were reported as major constraints followed by lack of inputs and no access to land (Figure 5).

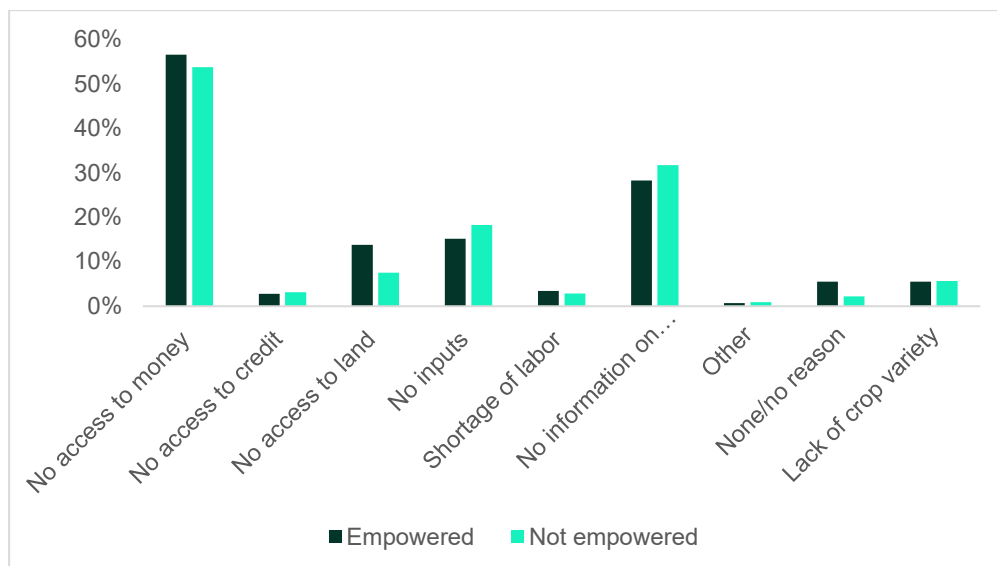


Figure 5: Key constraints reported by participants who are empowered and not empowered (source: Pro-WEAI analysis)

This aligns with insights from the qualitative findings. While interview participants mentioned recent community initiatives, such as tree planting, which aim to protect against soil erosion and conserve water, they also admitted they felt largely unprepared due to inadequate resources and technical knowledge. As shared by a farmer who planned to practice drip irrigation and change primary income source in anticipation of future shocks:

“Preparing for risks is hard because it is unknown... Having piped water would help, we don’t have that currently” (man, 57 years old, Kosirai).

Although some households have taken modest measures, such as digging trenches around homes to protect against floods, and digging boreholes near alternative water sources like springs, most lack comprehensive plans or access to climate information to prepare in advance. As narrated by the same farmer who keeps livestock and grows avocados, this “unpredictability is costing agriculture. We no longer have proper timing. It is hard to plan ahead” (man, 57 years old, Kosirai).

This demonstrates concerns around anticipatory capacities for early action. Some participants highlighted collective strategies that helped prepare them for shocks, such as jointly investing in tree nurseries and coming together to dig trenches for water management. In addition, sharing information with family members, neighbors, and friends was acknowledged as important for adaptive capacities.

A chi-squared test of independence revealed there is association between the two groups of only livestock keepers and livestock keepers who are also tea and horticulture farmers ($X^2=18.9582$, 1 df, $p<0.001$) for the following coping strategies: 1. did nothing, and 2. bought food/changed eating patterns (Table 11).

Table 11: Coping strategies reported by value chain groups

	Livestock only (n=330)	Tea + other horticulture farming and Livestock (n=329)	All
Did nothing ***	63%	79%	71%
Sale of assets	13%	8%	10%
Informal/formal credit	4%	1%	2%
Food relief	1%	0%	0%
Migrated	2%	1%	2%
Bought food/changed eating patterns ***	22%	11%	17%
Technology change (c and l)	7%	5%	6%
Other [please specify]	8%	7%	7%

Changing eating patterns or buying food as a coping strategy could have negative implications for women's nutrition and health status, as women are more likely to reduce their consumption, allocate more nutritious diets to others in the household, or use their savings to take care of food-related basic needs during times of shocks.

For livestock keepers also engaged in tea and horticulture, a chi-squared test revealed statistically significant association between climate-related shocks and changes in farming practices ($\chi^2 = 15.64$, $df = 4$, $p < 0.05$). They are likely not to change their practices, especially in the short term. This could be indicative of an already diversified portfolio involving both crops and livestock in a mixed crop-livestock system that can help buffer them against immediate effects of the shocks.

Additionally, for this group of livestock keepers who are also tea and horticulture farmers, it may not be easy to completely change farming practices. For example, adopting crop-related measures like cultivating a different type of crop on a land suitable for growing tea would require adequate access to capital and resources including information and technology. Lack of access to capital, information and inputs were reported as key constraints, even by those who are empowered, in response to the question of why they did not change their farming practices.

4. Discussion

This study provides a snapshot of empowerment dynamics and their relationship to resilience capacities among smallholder farming households in Nandi County, Kenya. Analysis using Pro-WEAI that provides a standardized and multidimensional measure of empowerment reveals **women and men are similarly disempowered**. Only 9% of women and 18% of men met the 80% adequacy threshold. The average 3DE score was higher for men (0.67) than for women (0.61), while the Gender Parity Index (GPI) stood at 0.87, reflecting a gender gap in empowerment within households. The overall Pro-WEAI score was 0.64. Our analysis further revealed inadequacy in indicators for both women and men across multiple domains of agency. Intra-household dynamics are not equitable, with men achieving greater adequacy compared to women in more than half of the households. Parity was achieved in less than one fourth of the households.

Qualitative data collected through FGDs and interviews complemented findings from the quantitative analysis by providing **“emic” perspectives on empowerment**. Women associated empowerment with independence, financial stability and decision-making, while men highlighted economic benefits and ability to support others. However, empowerment was not universally perceived as positive. Both women and men raised concerns about backlash and envy, associated with being seen as “empowered.” Notably, participants from the qualitative study did not consider themselves empowered, citing lack of capital and access to resources.

Inadequacy by indicators reveals that **contributors to disempowerment are mainly concentrated within domains of intrinsic and instrumental agency**. Overall, inadequacy was observed across indicators of self-efficacy, autonomy in income, and control over use of income, with women having greater inadequacy in work balance and visiting important locations compared to men. Such gender differences in inadequacy may be indicative of women’s time poverty due to unequal distribution of unpaid domestic and caregiving responsibilities, along with restrictive norms around women’s mobility. Both the burden of work and associated lack of time as well as restricted freedom of movement can then limit women’s ability to leave their homestead and engage in group meetings (Malapit et al., 2019).

Groups can act as vital platforms for access to knowledge, finance, and networks through pooling of resources, especially for women. Interventions often use groups to build social capital and train participants (Malapit et al., 2019). Relative to other indicators, women and men demonstrated **adequacy in group membership as an indicator of collective agency**. This aligns with insights from the qualitative data on women participating in table banking groups. Membership in these groups may also help buffer against economic and climate shocks. In comparison, the indicator membership in influential groups is a key contributor to disempowerment for both women and men, suggesting that while participation in collectives is common, influence within decision-making spaces may remain limited.

Qualitative data from the focus groups supports this. While the groups mentioned by women participants, specifically informal groups, may offer multiple benefits, they often face challenges related to management and lack of access to formal finance. Economic constraints, reported by both women and men, emerged as a central theme from the qualitative findings. However, while for men it is about inadequate capital and resources, for women, it is not merely the lack of access to finance, but also inability to have control over the use of financial resources, including income and savings. This is underpinned by cultural norms and traditional household structures that continue to place men in dominant decision-making roles (Bullock et al., 2025).

Some inconsistencies between quantitative and qualitative data are noteworthy. For instance, the Pro-WEAI results show low inadequacy in ownership of land and other assets and access to and decisions on financial services for both women and men, yet qualitative data emphasized lack of access to land, capital, and formal credit as critical barriers to empowerment. These divergences underscore the importance of mixed method approaches to capture context-specific nuances that may be obscured by standardized metrics.

Measures of empowerment varied only slightly by value chain, with notable similarities. For instance, intra-household gender gaps persisted across both groups, with men consistently demonstrating greater adequacy. This indicates that while livelihood

configuration may influence empowerment dimensions and gender parity, key role played by gender norms in shaping resource access and agency in a particular context cannot be ignored and must be integrated in our understanding of barriers to empowerment.

We further explored how empowerment relates with capacities to cope and adapt in the face of climate-related shocks and stresses such as droughts, floods, and erratic rainfall. A chi-squared test revealed significant association between empowerment and whether survey participants changed their farming practices in response to these shocks. **Those who are empowered are less likely to change their practices.** This may indicate greater absorptive capacity to buffer from immediate effects of shocks. Similar findings emerged for livestock keepers who also grow tea and horticultural crops. Their already diversified portfolio in a mixed crop-livestock system may have served as an internal cushion, reducing the urgency to adapt practices immediately in the aftermath of a shock. This is supported by literature that shows how capacity to buffer can depend on the ability to diversify at farm level and in livelihood strategies by spreading risk (e.g., see Matter et al., 2021).

Not changing practices or “doing nothing” as a response to shocks may also encompass a range of mundane actions that are part of people’s everyday adaptation (Castro and Sen, 2022). Perceptions or awareness of climate impacts may not always translate into action. Empirical evidence from other contexts suggests that a part of this puzzle of “doing nothing” to adapt can be connected to an invisibility surrounding local strategies that often go unrecognized as adaptation practices. In addition, those who say they are doing nothing may be expressing a sense of futility or powerlessness—a psychological barrier that can help explain the disconnect between knowing and doing (see Sharan, 2025).

Nonetheless, “doing nothing” may also point to barriers. In the context of this study, these barriers are faced by both empowered and disempowered groups. Lack of capital, input, or information were reported as key constraints to changing practices, even by those who are empowered. In such cases, empowerment may not be sufficient to drive behavioral change without addressing and changing restrictive norms and practices as root causes underpinning women’s limited agency in decision-making around uptake of adaptation innovations, and inequitable access to financial services, extension, and climate information.

Our study is not without limitations. We used a cross-sectional design, collecting data at a single point in time. As such, the findings reflect a snapshot rather than capturing changes over time or the impact of a specific intervention. Additionally, while questions related to climate shocks and reactive practices were integrated into the survey module, their placement and depth may have limited the precision with which resilience capacities, explored more within the qualitative data, could be linked to empowerment domains. Another limitation lies in the level of integration between the quantitative and qualitative components. While mixed methods were used, greater alignment in the design and sequencing of instruments could have strengthened triangulation and interpretation. We also acknowledge the inherent challenges in measuring empowerment quantitatively, especially in capturing relational and process-based dimensions of agency. These constraints underscore the importance of mixed methods and highlight the need for longitudinal data collection efforts to track empowerment and resilience processes over time.

Overall, our findings underscore that empowerment in this context is multidimensional and relational. The integration of emic perspectives is essential for understanding how empowerment is lived, experienced, and constrained—often in ways that standardized

measures may not fully capture. The relationship between empowerment and resilience is not linear and unidirectional. Empowerment may not only be a crucial pathway to strengthen resilience capacities of households in the face of shocks, but improved resilience capacities may also be a pathway to empowerment. Under certain conditions, empowerment can enhance absorptive capacity. However, future research using longitudinal data and mixed method design would be important for a deeper probing of the relationship between empowerment and climate resilience, more specifically to understand how empowerment can strengthen capacities to anticipate and adapt in the long run, and how enhancing these capacities can in turn serve as pathways to empowerment.

5. Conclusions and recommendations

This study offers a multidimensional assessment of empowerment and its relationship to climate resilience among smallholder farmers in Nandi County, Kenya. Using Pro-WEAI, we found that only a small fraction of the population is empowered—9% of women and 18% of men. Indicators across all three domains of agency contribute to disempowerment. Intra-household disparities persist, with more than 50% of households showing higher adequacy scores for men. Qualitative findings complemented insights from the quantitative analysis, underscoring empowerment as relational and context-specific, and extending beyond individual attributes to reflect community level dynamics.

The findings further suggest that empowerment is associated with responses to climate shocks. Those empowered were more likely to report “doing nothing” in the face of shocks—not necessarily out of inaction, but potentially due to their capacity to absorb impacts through diversification or reliance on existing strategies. However, structural constraints such as lack of capital, inputs, and information act as barriers for both empowered and disempowered groups, highlighting that empowerment alone is insufficient without enabling systems and more equitable norms and practices shaping resource access and decision-making.

Mixed methods revealed convergences between quantitative and qualitative findings. For instance, lower inadequacy in group membership as an indicator of collective agency is supported by qualitative findings. However, while the Pro-WEAI indicates lower inadequacy in ownership of land and other assets and access to and decisions on financial services, qualitative data emphasize these as persistent barriers. This divergence reinforces the need for complementary and in-depth qualitative inquiry to capture context-specific nuances and complexities.

Empowerment and resilience are not unidirectional. Rather, they are interlinked processes that can reinforce one another. Empowerment can enhance households' capacities to cope with climate impacts, but it may also result from strengthened resilience capacities that build agency and confidence to make timely decisions. Understanding this dynamic relationship requires continued investment in research integrating longitudinal mixed methods design. Ultimately, these findings call for tailored, inclusive interventions that go beyond individual empowerment to challenge structural inequities, support everyday adaptive strategies, and recognize the critical role of gender norms and practices in shaping outcomes. Key recommendations for research and programming are summarized below:

Address structural barriers: Programs should go beyond individual empowerment to address systemic constraints—such as inequitable access to finance, inputs, and climate information—that limit women’s and men’s ability to decide and act. Unpacking how these barriers intersect with empowerment and resilience would be important, especially in mixed crop-livestock systems.

Engage both women and men: Program design should actively include men and address intra-household dynamics, ensuring that efforts to empower women do not trigger unintended consequences like perceived disempowerment among men resulting in potential backlash for women.

Elevate everyday informal practices and strategies: Recognize and support everyday informal practices to cope with and adapt to climate impacts as forms of resilience. Research should deepen understanding of how these informal strategies contribute to absorptive, anticipatory, and adaptive capacities, particularly in relation to women’s empowerment.

Invest in longitudinal mixed methods approaches: Future research should move beyond snapshots in time to focus on causal mechanisms and metrics assessing long term impact using longitudinal, mixed methods design.

Advance understanding of transformative capacity: To advance understanding of transformative capacity, there is need to understand potential transformative impact of solutions that already exist within communities. Moving the lens from externally driven technical fixes towards positive deviant practices that depart from restrictive norms (e.g., see Bullock et al., 2025) can serve as entry points to strengthen links between empowerment dimensions and equitable and transformative resilience outcomes.

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Annex 1

Table 1: Demographic characteristics of survey participants in the overall sample

		Mixed household		Female only household
Characteristics		Male	Female	Female
Age	Mean (SD)	50.101(14.25)	46.694(12.49)	46.065(13.13)
Marital status	Currently married	90%	90%	55%
	Never married	8%	6%	6%
	Separated/deserted	0%	4%	6%
	Widow/widower	2%	0%	32%
	living together as married			
Education	Can read and write	96%	98%	97%
	Can read only	0%	0%	0%
	Can sign / write only	3%	1%	0%
	Cannot read and write	1%	1%	3%
	Never attended school	2%	5%	3%
	Primary	31%	40%	35%
	Junior Secondary (JSS)	1%	3%	3%
	Senior Secondary	38%	33%	35%
	College (Diploma, TVET)	22%	13%	23%
	Higher education (Degree, masters, etc.)	5%	5%	0%
Occupation	Non-earning occupation	2%	4%	3%
	Other self-employment	1%	0%	0%
	Salaried worker	5%	7%	3%
	Self-employed (farmer)	75%	76%	81%
	Self-employed (livestock & poultry se..	0%	1%	0%
	Self-employed (production)	0%	0%	0%
	Self-employed (professional)	3%	2%	0%
	Self-employed (trader)	5%	4%	10%
	Wage labour	4%	2%	3%
	not applicable	4%	3%	0%
Household size	Mean (SD)	5.28(1.87)		4.097(1.7001)

Table 2: Demographic characteristics of survey participants by value chains

		Livestock			Livestock, tea and horticulture			Grand Total
		Female only HH	Mixed household		Female only HH	Mixed household		
		Women	Women	Men	Women	Women	Men	
Age	Mean		46.3	48.	47.	47.2	51.	
		43	0	17	74	9	85	
		.69						
Household size	Mean		5.4	5.	3.	5.0	5.	
		4	6	46	74	7	12	
		.38						
Education	Can read and write	12	113	105	18	94	101	443
	Can read only		1					1
	Can sign / write only			4		2	2	8
	Cannot read and write	1	1	2		1		5
	Total	13	115	111	18	97	103	457
	Never attended school	1	5	3		4	2	15
	Primary	4	51	32	7	34	35	163
	Junior Secondary (JSS)		2	3	1	5		11
	Senior Secondary	4	36	44	7	34	37	162
	College (Diploma, TVET)	4	14	23	3	15	23	82
	Higher education (Degree, masters, etc.)		7	6		5	6	24
	Total	13	115	111	18	97	103	457
Occupation	Non-earning occupation		4	5	1	4		14
	Other self-employment			2				2
	Salaried worker	1	6	9		9	2	27
	Self-employed (farmer)	10	89	78	15	73	82	347
	Self-employed (livestock & poultry se..		1			1		2
	Self-employed (production)		1					1
	Self-employed (professional)		3	2		2	4	11
	Self-employed (trader)	1	3	6	2	6	6	24
	Wage labor	1	3	4		1	5	14
	not applicable		5	5		1	4	15
	Total	13	115	111	18	97	103	457

Annex 2

FGD Guide: Capacities for Innovation

Module 1 Local perceptions on empowerment and community agency trends (1.5 hours)

Perceptions and experiences surrounding empowerment (30 mins)

1. What qualities does someone you look up to have? What is he/she like? What is her/his life like? What have they achieved, or are able to do?
2. What can help a woman to be more admired within her household? Within her community?
3. Is it a good thing for a woman to be looked up to/admired? Why and why not? What are advantages? What are disadvantages? (probe for household and for community-level)
4. What are some problems women may have if they wish to earn income in this community? (probe regarding whether these issues are related to gender norms, the gender division of labour (available time); decision making power etc).
5. What are some problems women may have if they want to make important decisions about the household?
6. What have you heard about 'empowerment'? Is this different than what you think about for an admired person?
7. What do you think it means? What are elements of empowerment?
8. How are empowered women regarded? By other women? By other men? By their community? Has this changed/improved over time?
9. What do you hope for your daughter and son in their marriage? How would you like them to live?

Module 2 Local practices and innovative capacity (2 hours)

1. We'd now like to focus on new local practices surrounding agriculture and managing natural resources like the forests, water, and soils which surround this village. Thinking back over the past five years or so, what cropping or livestock practices or ways of managing local natural resources or organizing agricultural activities have people here tried out or experimented with? What are you doing differently than 5-10 years ago? Why? What are the benefits of the new practice?
2. Which of these new agricultural practices have been the most important for the [sex of FGD] in the community? Why important?
3. And which have been most important for the [opposite sex of FGD]? Why? [Please rank the top 2 in order of importance for each sex, and be sure to get explanations for the ratings.] What about for other marginalized groups in the community? Who are these? Poor? Of a certain religion? Or a different ethnic group?

New agricultural or NRM practices	Most important for the [sex of the FGD] of the community (top 2)?	Most important for the [opposite sex of FGD] of the community (top 2)

4. How did you, or members in the community, come to learn about the new approach/technology? Was it your neighbors/peers, government extension, a local NGO, international researchers? How did/does information you receive from these actors differ?
5. Do you also share information about the new approach/technology? How and with whom?
6. Who is learning from agricultural extension agents? Only men or also women? What other groups? Why or why not?
7. Was [THE NUMBER ONE RANKED NEW PRACTICE FOR FGD SEX] or elements of this practice, rapidly adopted by local people, or were there any problems with its uptake? What elements were adopted, and which weren't? [Pause for responses.]
8. Now we're going to explore in greater detail [THE NEW PRACTICE]. Tell me about who were the first to learn about and try out this new practice.
9. Have any local men or women tried out new ways to make [THE NEW PRACTICE] work better here? [If so] Please tell me about this. [Probe very deeply on this.]
10. What are people like who are always trying out or inventing new ways of doing something? What kind of personality do they have?
11. In your household or experience in the village/community, who is more inclined to change things/improve how you farm? A man or woman? Why?
12. Are there any formal or informal groups of people who are working together on [THE NEW PRACTICE]? If so, tell me about the group and its members and activities. [Also probe deeply on this, including whether these networks are mainly men or women or both.]

13. We just spoke about people who change things in agricultural production – maybe ourselves or others. Let's step back now and think about what helps people here to try things out and change practices? How do you decide to use them or not (e.g. new varieties)?

14. What kinds of resources (money, labour, information) are available in the village that enable men and women to learn about and try—or possibly create themselves—something new that could make their livelihoods more productive or that could improve the wellbeing of their families? What resources are available for men, that are not for women?

15. What are the problems that may lead to farmers not adopting? What do you think about the cost, time and labor needs it requires? What is the accessibility (is there a location closeby where you can buy seeds/inputs)? Is there a market or demand locally for the product?

16. Among the factors that support innovation, which two do you think are the most important for the [sex of FGD] of the community? Why? [Create the second column, and indicate the 1st and 2 nd second most important factors.] D.15 Now, let's think about barriers. What are the factors that hinder local people from trying out innovations? And of these, which two are the most important barriers for the [sex of FGD]?

	Two most important for the [sex of FGD]
Factors that support innovation	
Factors that hinder innovation	

17. Let's say you have a plot available; how do you decide in your HH what to plant? Whether to use inputs? Can you decide this alone?

18. Imagine it's time to pay school fees and buy uniforms, but you also need inputs on your farm. How do you decide what to prioritize? And can you make this decision alone?

Demographic characteristics of FGD participants

FGD 1: Kaptumo, Women

ID	Age	Marital status	Ethno-religious group	Primary occupation	Secondary occupation	Level of education completed	Length of time lived in the community (years)
R1	58	Married	Christian	Farmer	NA	Primary	30
R2	60	Married	Christian	Farmer	Business	Form 4	35
R3	48	Single	Christian	Farmer	NA	Form 4	20
R4	60	Widow	Christian	Farmer	NA	Class 7	38
R5	45	Married	Christian	Farmer	Business	Form 4	45
R6	47	Married	Christian	Farmer	Business	Form 3	47

FGD 2: Kapkangani, Men

ID	Age	Marital status	Ethno-religious group	Primary occupation	Secondary occupation	Level of education completed	Length of time lived in the community (years)
R7	52	Married	Christian	Farmer	NA	Primary	52
R8	71	Married	Christian	Farmer	NA	O Level	71
R9	60	Married	Christian	Farmer	NA	O Level	60
R10	57	Married	Islam	Farmer	NA	O Level	57
R11	62	Single	Christian	Farmer	NA	College	28
R12	63	Married	Christian	Farmer	NA	Form 4	50
R13	47	Married	Christian	Farmer	NA	College	47
R14	46	Married	Christian	Farmer	Business	College	46

FGD 3: Kosirai, Men

ID	Age	Marital status	Ethno-religious group	Primary occupation	Secondary occupation	Level of education completed	Length of time lived in the community (years)
R15	70	Married	Christian	Farmer	NA	College	70
R16	49	Married	Christian	Farmer	Business	O Level	49
R17	43	Married	Christian	Farmer	NA	College	40
R18	46	Married	Christian	Farmer	NA	Primary	46
R19	35	Married	Christian	Farmer	NA	Primary	35
R20	40	Married	Christian	Farmer	NA	College	35
R21	38	Married	Christian	Farmer	NA	O Level	38

FGD 4: Kipkaren, Women

ID	Age	Marital status	Ethno-religious group	Primary occupation	Secondary occupation	Level of education completed	Length of time lived in the community (years)
R22	36	Married	Christian	Farmer	NA	College	20
R23	45	Married	Christian	Farmer	Employed	Form 3	25
R24	59	Married	Christian	Farmer	NA	Form 4	35
R25	32	Married	Christian	Farmer	NA	Class 7	16
R26	38	Single	Christian	Farmer	Self-employed (grocery)	Class 8	38
R27	59	Married	Christian	Farmer	NA	Form 4	39
R28	71	Married	Christian	Farmer	NA	Form 4	57
R29	30	Married	Christian	Farmer	NA	Form 2	10

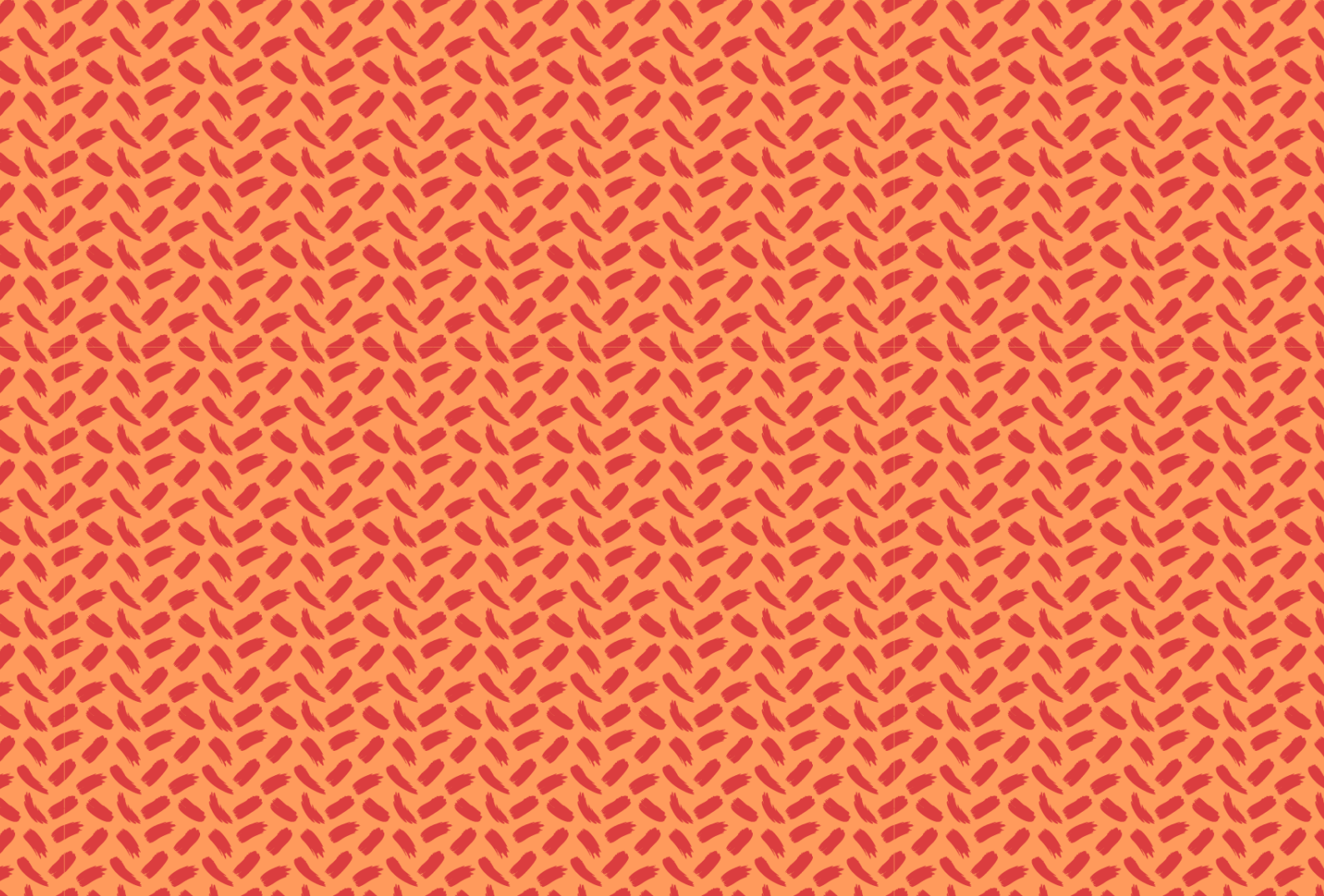


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